

Decomposition methods for large-scale convex and non-convex problems

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2 A recap on convex analysis

- Main tools
- The Fenchel-Rockafellar duality theorem

- Frank-Wolfe algorithm
- The non-convex case
- Dual Subgradient

4 Proximal methods

The aggregative problem:

$$\inf_{x=(x_1, \dots, x_N)} P(x) := \frac{1}{N} \sum_{i=1}^N F_i(x_i) + G\left(\frac{1}{N} \sum_{i=1}^N A_i x_i\right), \quad (\mathfrak{P})$$

- $F_i: \mathbb{R}^{n_i} \rightarrow \bar{\mathbb{R}}$ (the individual costs)
- $A_i \in \mathbb{R}^{m \times n_i}$
- $G: \mathbb{R}^m \rightarrow \bar{\mathbb{R}}$ (the coupling cost).

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Motivation

Energy management:

- N **flexible** production or consumption units, indexed by i
 - x_i denotes all operational decisions related to the unit i , $F_i(x_i)$ is the cost specific to i
 - $\frac{1}{N} \sum_{i=1}^N A_i x_i$ is an aggregative quantity, for example the total energy production/consumption
 - g may model:
 - a penalty cost for the distance to a target consumption profile
 - constraints, for example the maximum total consumption.
- See the **smart charging** problem proposed by Benjamin Dubois-Taine in his tutorial.

- each i refers to a random event of probability $1/N$
- x_i is the decision in the event i , with cost $F_i(x_i)$
- G penalizes an “expected quantity”.

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- ... see the dual problem!

Numerical resolution

Focus on **decomposition methods** (a.k.a. splitting methods): each iteration requires the solving of (atmost) N independant subproblems, associated with each index i .

Advantages:

- parallelizability
- scalability as N increases.

An essential feature: the Lagrangian of the problem is **separable**.

An old story! Focus on **modern primal-dual algorithms**.

- the convex case: all F_i and G are convex
- the non-convex case: only G is convex.

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- the smooth case: G is differentiable with Lipschitz-continuous gradient
- the non-smooth case: any other situation.

We will always assume that the F_i have a bounded domain.

Remarks

- Do we care about the $\frac{1}{N}$ -factor? Yes... and no.
- Capital letters for the aggregative problem, small letters for the generic problems.

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- TABLE 1. *Continued*

Fenchel conjugate

Let $f: \mathbb{R}^n \rightarrow \bar{\mathbb{R}}$. Let $\lambda \in \mathbb{R}^n$ and let $z \in \mathbb{R}$. The affine function $x \mapsto -z + \langle \lambda, x \rangle$ is a lower bound of f if

$$f(x) \geq -z + \langle \lambda, x \rangle, \forall x \in \mathbb{R}^n \iff z \geq \sup_{x \in \mathbb{R}^n} \langle \lambda, x \rangle - f(x)$$

We call **Fenchel conjugate** the function defined by

$$f^*(\lambda) = \sup_{x \in \mathbb{R}^n} \langle \lambda, x \rangle - f(x) \in \mathbb{R} \cup \{-\infty, +\infty\}.$$

When $f^*(\lambda)$ is finite, then $x \mapsto -f^*(\lambda) + \langle \lambda, x \rangle$ is a **tangent line** of the graph of f .

Lemma

For any $f \in \Gamma_0(\mathbb{R}^n)$, $f^* \in \Gamma_0(\mathbb{R}^n)$.

Duality theorem

Theorem (Fenchel-Moreau)

Let $f \in \Gamma_0(\mathbb{R}^n)$. Then f can be represented as the supremum of all its tangent lines:

$$f(x) = \sup_{\lambda \in \mathbb{R}^n} \langle \lambda, x \rangle - f^*(\lambda).$$

In short, $f = (f^)^*$.*

Subdifferential

Let $f \in \Gamma_0(\mathbb{R}^n)$ and let $x \in \text{dom}(f)$. The **subdifferential** of f at x , denoted $\partial f(x)$ is defined as the set of vectors λ such that

$$f(x') \geq f(x) + \langle \lambda, x' - x \rangle, \quad \forall x' \in \mathbb{R}.$$

Lemma (Fenchel-Young)

Let $f \in \Gamma_0(\mathbb{R}^n)$. Let x and λ in $\mathbb{R}^n$. Then

$$\begin{aligned}\lambda \in \partial f(x) &\iff f(x) + f^*(\lambda) - \langle \lambda, x \rangle = 0 \\ &\iff f^{**}(x) + f^*(\lambda) - \langle \lambda, x \rangle = 0 \iff x \in \partial f^*(\lambda).\end{aligned}$$

A **very** useful corollary is the following:

$$\partial f^*(\lambda) = \operatorname{argmin}_{x \in \mathbb{R}^n} f(x) - \langle \lambda, x \rangle.$$

Smoothness and strong convexity

Let $f: \mathbb{R}^n \rightarrow \bar{\mathbb{R}}$. We say that...

- f is **L -smooth** if f is differentiable and ∇f is L -Lipschitz continuous, i.e.,

$$\|\nabla f(x') - \nabla f(x)\| \leq L\|x' - x\|, \quad \forall x, x' \in \mathbb{R}^n$$

- α -**strongly** convex if $f(\cdot) - \frac{\alpha}{2}\|\cdot\|^2$ is convex.

Proposition

Let $f \in \Gamma_0(\mathbb{R}^n)$. Let $L > 0$. Then f is L -smooth if and only if f^* is $\frac{1}{L}$ -strongly convex.

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Intuition

Let $f \in \Gamma_0(\mathbb{R}^n)$, let $g \in \Gamma_0(\mathbb{R}^m)$, and let $A \in \mathbb{R}^{m \times n}$. Consider the abstract problem:

$$\inf_{x \in \mathbb{R}^n} p(x) := f(x) + g(Ax). \quad (\text{p})$$

Let $\bar{x}$ denote a solution to the problem.

Suppose that f and g are smooth, then

$$0 = \nabla p(\bar{x}) = \nabla f(\bar{x}) + A^\top \nabla g(A\bar{x}).$$

Equivalently: there exists $\lambda \in \mathbb{R}^n$ such that

$$\lambda \in \partial g(A\bar{x}) \quad \text{and} \quad -A^\top \lambda \in \partial f(\bar{x}).$$

This statement remains true for non-differentiable f and g !

Statement

Theorem (Fenchel-Rockafellar theorem)

Let $f \in \Gamma_0(\mathbb{R}^n)$, let $g \in \Gamma_0(\mathbb{R}^m)$, and let $A \in \mathbb{R}^{m \times n}$. Assume that

$$0 \in \text{int}\left(\text{dom}(g) - A \text{dom}(f)\right).$$

Then the primal problem $\mathfrak{p}$ and the dual problem $\mathfrak{d}$ defined below have the same value:

$$\sup_{\lambda \in \mathbb{R}^m} d(\lambda) := -f^*(-A^\top \lambda) - g^*(\lambda). \quad (\mathfrak{d})$$

In addition, if this value is finite, then the dual problem has a solution.

Min-max duality

Define the **Lagrangian** $\ell: \mathbb{R}^n \times \mathbb{R}^m \mapsto \mathbb{R} \cup \{-\infty, +\infty\}$ as

$$\ell(x, \lambda) = f(x) + \langle \lambda, Ax \rangle - g^*(\lambda).$$

For the primal cost, we have

$$\begin{aligned} p(x) &= f(x) + g(Ax) \\ &= f(x) + \sup_{\lambda \in \mathbb{R}^m} \langle \lambda, Ax \rangle - g^*(\lambda) \\ &= \sup_{\lambda \in \mathbb{R}^m} \ell(x, \lambda). \end{aligned}$$

Similarly, $d(\lambda) = \inf_{x \in \mathbb{R}^n} \ell(x, \lambda)$.

As a consequence:

$$\inf_{x \in \mathbb{R}^n} \sup_{\lambda \in \mathbb{R}^m} \ell(x, \lambda) = \sup_{\lambda \in \mathbb{R}^m} \inf_{x \in \mathbb{R}^n} \ell(x, \lambda).$$

Optimality conditions

Corollary

Let $(x, \lambda) \in \mathbb{R}^n \times \mathbb{R}^m$. Then, the following holds:

x is a solution to (p), λ is a solution to (d)

$$\iff -A^\top \lambda \in \partial f(x) \text{ and } \lambda \in \partial g(Ax)$$

$$\iff x \text{ minimizes } \ell(\cdot, \lambda) \text{ and } \lambda \text{ maximizes } \ell(x, \cdot).$$

Warning! Fix λ a solution to the dual problem. Then the following inclusion may be strict:

$$\operatorname{argmin}_{x \in \mathbb{R}^n} p(x) \subset \operatorname{argmin}_{x \in \mathbb{R}^n} \ell(\cdot, \lambda).$$

If x minimizes $\ell(\cdot, \lambda)$, then x may not be a solution to (p).

Application to (P)

Recall the aggregative problem of interest:

$$\inf_{x=(x_1, \dots, x_N)} P(x) := \frac{1}{N} \sum_{i=1}^N F_i(x_i) + G\left(\frac{1}{N} \sum_{i=1}^N A_i x_i\right). \quad (\mathfrak{P})$$

For this problem, the Lagrangian writes:

$$L(x, \lambda) = \frac{1}{N} \sum_{i=1}^N \left(F_i(x_i) + \langle \lambda, A_i x_i \rangle \right) - G^*(\lambda).$$

It is **separable** with respect to the variables x_i . The dual cost is:

$$D(\lambda) = -\frac{1}{N} \sum_{i=1}^N F_i^*(-A_i^\top \lambda) - G^*(\lambda).$$

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Algorithms and linear oracle

Back to the aggregative problem. We investigate two classes of methods, for the following situations:

- the **Frank-Wolfe** algorithm (a.k.a. conditional gradient), when G is L -smooth
- the **dual subgradient** algorithm, when $G = \iota_b$, i.e., $G(y) = 0$ if $y = b$ and $G(y) = +\infty$, otherwise.

Common denominator. Both methods require to be able to solve the following **subproblems**, for $i = 1, \dots, N$:

$$\inf_{x_i \in \mathbb{R}^{n_i}} F_i(x_i) + \langle A_i^\top \lambda, x_i \rangle.$$

In the complexity estimates to be realized, “one call to the oracle” means “solving the subproblem, for one value of λ and i ”.

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Application to the aggregative problem

We investigate the resolution with the Frank-Wolfe algorithm of the aggregative problem P :

$$\inf_{x=(x_1, \dots, x_N)} P(x) := \frac{1}{N} \sum_{i=1}^N F_i(x_i) + G\left(\frac{1}{N} \sum_{i=1}^N A_i x_i\right), \quad (\mathfrak{P})$$

under the following assumptions:

- for all $i = 1, \dots, N$, $F_i \in \Gamma_0(\mathbb{R}^{n_i})$, $G \in \Gamma_0(\mathbb{R}^m)$
- for all i , F_i has bounded domain
- G is L -smooth.

We denote as D a bound on the diameters of $\text{dom}(F_i)$, $\|A\|$ a bound of $\max_i(\|A_i\|)$, supposed independent of N .

Abstract setting

Abstract problem with $f \in \Gamma_0(\mathbb{R}^n)$ and $g \in \Gamma_0(\mathbb{R}^m)$:

$$\inf_{x \in \mathbb{R}^n} p(x) := f(x) + g(Ax), \quad (\text{p})$$

Assumptions:

- f has bounded domain, with diameter D
- g is L -smooth.

Oracle: for any $\lambda \in \mathbb{R}^m$, one can solve the subproblem (a.k.a. linearized problem)

$$\inf_{s \in \mathbb{R}^n} f(s) + \langle A^\top \lambda, s \rangle.$$

Two observations:

- This amounts to minimizing $\ell(\cdot, \lambda)$.
- If $\lambda = \nabla g(Ax)$, then $\langle A^\top \lambda, s - x \rangle \approx g(As) - g(Ax)$.

Frank-Wolfe algorithm

Algorithm 1: Frank-Wolfe algorithm for (p)

Initialize: $x^0 \in \text{dom}(f)$;

for $k = 0, 1, \dots$ **do**

 Set $\lambda = \nabla g(Ax^k)$;

 Find $s^k \in \text{argmin } f(\cdot) + \langle A^\top \lambda^k, \cdot \rangle$;

 Choose $\theta_k \in [0, 1]$;

 Set $x^{k+1} = (1 - \theta_k)x^k + \theta_k s^k$;

end

Theorem

Let $\bar{x}$ denote a solution to (p). Choose $\theta_k = \frac{2}{k+2}$, $\forall k \in \mathbb{N}$. Then,

$$p(x^k) - p(\bar{x}) = \mathcal{O}\left(\frac{1}{k}\right).$$

Elements of proof

For all $k \in \mathbb{N}$, consider the **primal-dual gap**:

$$\Delta_k = \left(f(x^k) + \langle A^\top \lambda^k, x^k \rangle \right) - \left(f(s^k) + \langle A^\top \lambda^k, s^k \rangle \right).$$

One can show the following:

- Certificate: $\Delta_k = p(x^k) - d(\lambda^k) \geq \gamma_k := p(x^k) - p(\bar{x}).$
- Decay: $p(x^{k+1}) \leq p(x^k) - \theta_k \Delta_k + \frac{1}{2} L D^2 \|A\|^2 \theta_k^2.$

Combining the two, we obtain:

$$\gamma_{k+1} \leq (1 - \theta_k) \gamma_k + \frac{1}{2} L D^2 \|A\|^2 \theta_k^2.$$

An explicit upper bound of γ_k can be deduced.

Frank-Wolfe algorithm

Algorithm 2: Frank-Wolfe algorithm for $(\mathfrak{P})$

Initialize: $x^0 = (x_1^0, \dots, x_N^0) \in \prod_{i=1}^N \text{dom}(F_i)$;

for $k = 0, 1, \dots$ **do**

 Set $\lambda^k = \nabla G(\frac{1}{N} \sum_{i=1}^N A_i x_i^k)$;

for $i = 1, \dots, N$ **do**

 Find $s_i^k \in \text{argmin } F_i(\cdot) + \langle A_i^\top \lambda^k, \cdot \rangle$;

end

 Choose $\theta_k = 2/(k+2)$;

 Set $x^{k+1} = (1 - \theta_k)x^k + \theta_k s^k$;

end

Theorem

To achieve a precision of $\varepsilon > 0$, the algorithm requires $\mathcal{O}\left(\frac{N}{\varepsilon}\right)$ calls to the oracle.

Block-coordinate Frank-Wolfe algorithm

Technique. At any iteration k , instead of solving all subproblems $\inf F_i(\cdot) + \langle A_i^\top \lambda^k, \cdot \rangle$, just solve **one** of them, for a **randomly selected** index i .

Algorithm 3: Frank-Wolfe algorithm

Initialize: $x^0 = (x_1^0, \dots, x_N^0) \in \prod_{i=1}^N \text{dom}(F_i)$;**for** $k = 0, 1, \dots$ **do**Set $\lambda^k = \nabla G(\frac{1}{N} \sum_{i=1}^N A_i x_i^k)$;

Choose uniformly at random $I_k \in \{1, \dots, N\}$;

Find $s_{l_k}^k \in \operatorname{argmin} F_{l_k}(\cdot) + \langle A_{l_k}^\top \lambda^k, \cdot \rangle$;

Choose $\theta_k \in [0, 1]$;

For $i = l_k$, set $x_i^{k+1} = (1 - \theta_k)x_i^k + \theta_k s_i^k$;

Otherwise, set $x_i^{k+1} = x_i^k$;

end

Convergence

For problem $(\mathfrak{P})$, the primal-dual gap writes:

$$\Delta^k = \frac{1}{N} \sum_{i=1}^N \Delta_i^k, \quad \text{where:}$$

$$\Delta_i^k = \left(F_i(x_i^k) + \langle A_i^\top \lambda^k, x_i^k \rangle \right) - \left(F_i(s_i^k) + \langle A_i^\top \lambda^k, s_i^k \rangle \right).$$

“At iteration k ”, we have $\Delta^k = \mathbb{E}[\Delta_i^k]$.

We adapt the previously conducted analysis:

$$p(x^{k+1}) \leq p(x^k) - \frac{1}{N} \theta_k \Delta_i^k + \frac{1}{2N^2} LD^2 \|A\|^2 \theta_k^2.$$

Taking the expectation yields:

$$\mathbb{E}[p(x^{k+1})] \leq p(x^k) - \left(\frac{\theta_k}{N} \right) \Delta^k + \frac{1}{2} LD^2 \|A\|^2 \left(\frac{\theta_k}{N} \right)^2.$$

Convergence

We get the “same” inequality as before, with θ_k replaced by θ_k/N .
So the new stepsize should be N -times larger than previously!

Theorem

Take $\theta_k = \frac{2N}{k+2N}$, for all $k \in \mathbb{N}$. Then

$$\mathbb{E}[P(x^k) - P(\bar{x})] = \mathcal{O}\left(\frac{1}{K} + \frac{N^2}{K^2}\right).$$

In other words, to reach a precision of $\varepsilon > 0$ (in expectation), the algorithm requires $\mathcal{O}\left(\frac{1}{\varepsilon} + \frac{N}{\sqrt{\varepsilon}}\right)$ calls to the oracle.

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Setting

We consider the following assumptions:

- F_i has bounded and non-empty domain, is bounded on its domain.
- $G \in \Gamma_0(\mathbb{R}^m)$ and G is L -smooth.

We do **not** require **any**more convexity of F_i .

Our approach relies on

- 1 a convex approximation of the problem
- 2 an estimate of the approximation
- 3 a numerical method combining Frank-Wolfe (and its variants) and a sampling technique.

Convex approximation

Main idea: use random variables $X = (X_1, \dots, X_N)$ and consider

$$\tilde{P}(X) = \frac{1}{N} \sum_{i=1}^N \mathbb{E}[F_i(X_i)] + G\left(\frac{1}{N} \sum_{i=1}^N \mathbb{E}[A_i X_i]\right).$$

Warning: $\tilde{P}$ is not convex, and $\tilde{P}(X) \neq \mathbb{E}[P(X)]$.

We do not optimize the X_i directly: we optimize their **distributions** $\mu_i \in \mathcal{P}(\mathbb{R}^{n_i})$. The relaxed cost is defined, for $\mu = (\mu_1, \dots, \mu_N)$, as

$$J(\mu) = \frac{1}{N} \sum_{i=1}^N \int F_i(x_i) d\mu_i(x_i) + G\left(\frac{1}{N} \sum_{i=1}^N \int A_i x_i d\mu_i(x_i)\right).$$

So if $X_i \sim \mu_i$, then $\tilde{P}(X) = J(\mu)$. Now J is **convex** and has a suitable **aggregative structure**.

Quality approximation

Proposition

Let $X_1, \dots, X_N$ be independant random variables, distributed as $\mu_1, \dots, \mu_N$. Then it holds that:

$$\mathbb{E}[P(X)] \leq \tilde{P}(X) + \frac{L}{2N^2} \sum_{i=1}^N \text{Var}(A_i X_i) = \tilde{P}(X) + \mathcal{O}\left(\frac{1}{N}\right),$$

where $\text{Var}(A_i X_i) = \mathbb{E}\left[\|A_i X_i - \mathbb{E}[A_i X_i]\|^2\right]$.

We call **relaxed problem** the following:

$$\inf_{\mu \in \prod_{i=1}^N \mathcal{P}(\mathbb{R}^{n_i})} J(\mu). \quad (\tilde{\mathfrak{P}})$$

We have:

$$\text{Val}(\tilde{P}) \leq \text{Val}(P) \leq \text{Val}(\tilde{P}) + \frac{C}{N}.$$

Sampling

Lemma

Let $x^{(1)}, \dots, x^{(J)}$ be J realizations of X (as before). Then

$$\mathbb{P}\left[\min_{j=1,\dots,J} P(x^{(j)}) \leq J(\mu) + \frac{C}{N} + \delta\right] \geq 1 - \exp\left(-\frac{C'\delta^2}{Nn}\right).$$

Proof: apply McDiarmid's concentration inequality.

Practical consequence: given an ε -candidate μ for $(\tilde{P})$, we easily obtain an $(\varepsilon + C/N)$ -optimal candidate for (P) , through **sampling**.

Solving the relaxed problem

The relaxed problem still has an aggregative structure, with a separable Lagrangian. The subproblems (in the sense of Frank-Wolfe) write:

$$\begin{aligned} \inf_{\mu_i \in \mathbb{R}^{n_i}} \int F_i(x_i) d\mu_i + \left\langle \lambda^k, \int A_i x_i d\mu_i(x_i) \right\rangle \\ = \inf_{\mu_i \in \mathbb{R}^{n_i}} \int \left(F_i(x_i) + \langle A_i^\top \lambda^k, x_i \rangle \right) d\mu_i(x_i). \end{aligned}$$

Lemma

Let s_i^k denote a minimizer of $F_i(\cdot) + \langle A_i^\top \lambda^k, \cdot \rangle$. Then the Dirac mass centred at s_i^k is a solution to the subproblem.

In short: the Frank-Wolfe algorithm remains applicable in the relaxed setting. It relies on **exactly the same oracle** as originally.

Wrapping up the approach

Algorithm 4: Non-convex FWA with sampling

Initialize: $x^0 = (x_1^0, \dots, x_N^0) \in \prod_{i=1}^N \text{dom}(F_i)$;

Define $\mu^0 = (\delta_1^0, \dots, \delta_N^0)$;

for $k = 0, 1, \dots, K - 1$ **do**

 Set $\lambda^k = \nabla G(\frac{1}{N} \sum_{i=1}^N \int A_i x_i d\mu_i^k(x_i))$;

 Set $\theta_k = 2/(2 + k)$;

for $l = 1, \dots, N$ **do**

 Find $s_i^k \in \text{argmin } F_i(\cdot) + \langle A_i^\top \lambda^k, \cdot \rangle$;

 Set $\mu_i^{k+1} = (1 - \theta_k)\mu_i^k + \theta_k \delta_{s_i^k}$;

end

end

Compute samples $(x^{(j)})_{j=1, \dots, J}$ of μ^k ;

Find $j^* \in \text{argmin}_{j=1, \dots, J} P(x^{(j)})$;

Output: $x^{(j^*)}$.

Wrapping up the approach

Algorithm 5: Non-convex block-coordinate FWA with sampling

Initialize: $x^0 = (x_1^0, \dots, x_N^0) \in \prod_{i=1}^N \text{dom}(F_i)$;

Define $\mu^0 = (\delta_1^0, \dots, \delta_N^0)$;

for $k = 0, 1, \dots, K - 1$ **do**

Set $\lambda^k = \nabla G(\frac{1}{N} \sum_{i=1}^N \int A_i x_i d\mu_i^k(x_i))$;

Choose uniformly at random $l_k \in \{1, \dots, N\}$;

Find $s_{l_k}^k \in \text{argmin} F_{l_k}(\cdot) + \langle A_{l_k}^\top \lambda^k, \cdot \rangle$;

Set $\theta_k = 2N/(k + 2N)$;

For $i = l_k$, set $\mu_i^{k+1} = (1 - \theta_k)\mu_i^k + \theta_k \delta_{s_i^k}$;

Otherwise, set $\mu_i^{k+1} = \mu_i^k$;

end

Compute samples $(x^{(j)})_{j=1, \dots, J}$ of μ^k ;

Find $j^* \in \text{argmin}_{j=1, \dots, J} P(x^{(j)})$;

Output: $x^{(j^*)}$.

Mean-field relaxation

The relaxed problem is related to the concept of mean-field relaxation, when all agents are homogeneous, i.e.,

$$F_1 = \dots = F_N \quad \text{and} \quad A_1 = \dots = A_N.$$

The aggregative problem can be written as follows:

$$\inf_{\mu \in \mathcal{P}_N(\mathbb{R}^n)} \int F(x) d\mu(x) + G\left(\int Ax d\mu(x)\right),$$

$$\text{where: } \mathcal{P}_N(\mathbb{R}^n) = \left\{ \frac{1}{N} \sum_{i=1}^N \delta_{x_i} \mid (x_1, \dots, x_N) \in \mathbb{R}^{Nn} \right\}.$$

The relaxed problem writes exactly the same way, with $\mathcal{P}(\mathbb{R}^n)$ instead of $\mathcal{P}_N(\mathbb{R}^n)$.

In short: the relaxed problem can be interpreted as an **asymptotic problem** as $N \rightarrow \infty$ (in the case of homogeneous agents).

Comparison

In the non-convex framework:

- the oracle is the same, possibly is a non-convex problem
- we need to keep track of all outputs of the oracle
- no guarantee to reach a precision lower than $\mathcal{O}(1/N)$.

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Setting

We investigate the resolution with the **dual subgradient algorithm** of the aggregative problem with **coupling constraint**:

$$\inf_{x=(x_1,\dots,x_N)} P(x) := \frac{1}{N} \sum_{i=1}^N F_i(x_i), \quad \text{s.t.} \quad \frac{1}{N} \sum_{i=1}^N A_i x_i = b, \quad (\mathfrak{P})$$

assuming that

- $F_i \in \Gamma_0(\mathbb{R}^{n_i})$ has bounded domain
- a dual solution exists.

In this context, $G(y) = 0$ if $y = b$, $G(y) = \infty$ otherwise. We have $G^*(\lambda) = \langle \lambda, b \rangle$.

Oracle: for any $\lambda \in \mathbb{R}^m$ and $\omega \geq 0$, we must be able to solve

$$\inf_{x_i \in \mathbb{R}^{n_i}} \omega F(x_i) + \langle A^\top \lambda, x_i \rangle.$$

Dual subgradient

Consider the abstract optimization problem:

$$\inf f(x), \quad Ax = b \quad (\text{p})$$

where $f \in \Gamma_0(\mathbb{R}^n)$ has bounded domain.

The lagrangian is $\ell(x, \lambda) = f(x) + \langle \lambda, Ax - b \rangle$, the dual cost writes

$$d(\lambda) = \inf_{x \in \mathbb{R}^n} \ell(x, \lambda) = -f^*(A^\top \lambda).$$

The dual cost d is Lipschitz but not differentiable in general.

Lemma

For all λ , it holds that

$$\partial d(\lambda) = \left\{ y \mid y = Ax - b, x \in \operatorname{argmin} f(\cdot) + \langle A^\top \lambda, \cdot \rangle \right\}.$$

Dual subgradient

Algorithm 6: Dual subgradient for (p)

Initialize: $\lambda^0 \in \mathbb{R}^m$;

for $k = 0, 1, \dots, K - 1$ **do**

 Find $s^k \in \operatorname{argmin} f(\cdot) + \langle A^\top \lambda, \cdot \rangle$;

 Choose $\theta_k > 0$;

 Set $\lambda^{k+1} = \lambda^k + \theta_k (As^k - b)$;

end

Set $x^K = \frac{1}{K} \sum_{k=0}^{K-1} s^k$.

Convergence rate

The output x^K may not be feasible, i.e., possibly $Ax^K \neq b$. We use the following criterion to assess its optimality:

$$\text{opt}(x) = \left(f(x) - f(\bar{x}) \right)_+ + \|Ax - b\|,$$

where $\bar{x}$ denotes a solution to the problem and $(y)_+ := \max(y, 0)$.

Theorem

Assume the existence of a dual solution $\bar{\lambda}$. Fix a constant M and choose $\theta_k = \frac{M}{\sqrt{K}}$, for $k = 1, \dots, K$. Then there exists a constant C , independent of K , such that

$$d(\bar{\lambda}) - d(\lambda_K) \leq \frac{C}{\sqrt{K}} \quad \text{and} \quad \text{opt}(x^K) \leq \frac{C}{\sqrt{K}}.$$

Primal solutions

The dual subgradient algorithm comes up with a good primal candidate. This is not true anymore in the randomized version.

Suppose that you have found the exact value $d(\bar{\lambda})$ of the problem (but you do not know any primal solution).

Lemma

Consider the auxiliary problem:

$$\inf_{(x \in \mathbb{R}^n)} p_{\text{aux}}(x) := \frac{1}{2}(f(x) - d(\bar{\lambda}))_+^2 + \frac{1}{2}\|Ax - b\|^2. \quad (\mathbf{p}_{\text{aux}})$$

Its value is 0, moreover, it has the same solution set as (p).

Proof. Clearly $\text{Val}(\mathbf{p}_{\text{aux}}) \geq 0$. If x is a solution to (p), then $p_{\text{aux}}(x) = 0$, so x is a solution to $(\mathbf{p}_{\text{aux}})$. Conversely, if $p_{\text{aux}}(x) = 0$, then $f(x) = d(\bar{\lambda})$ and $Ax = b$, so x is a sol. to (p).

Solving the auxiliary problem

The auxiliary problem can be rewritten as:

$$\inf_{(x,z) \in \mathbb{R}^n \times \mathbb{R}} \frac{1}{2} (z - d(\bar{\lambda}))_+^2 + \frac{1}{2} \|Ax - b\|^2, \quad \text{subject to: } z \geq f(x).$$

One can solve it with the **Frank-Wolfe algorithm**. We fully linearize the cost function. At iterate k , given the candidate (x^k, z^k) , the subproblem writes:

$$\inf_{(x,z) \in \mathbb{R}^n \times \mathbb{R}} \omega^k z + \langle A^\top \lambda^k, x \rangle, \quad \text{subject to } z \geq f(x),$$

where $\omega^k = (z^k - d(\bar{\lambda}))_+$ and $\lambda^k = Ax^k - b$. This is equivalent to

$$\inf_{x \in \mathbb{R}^n} \omega^k f(x) + \langle A^\top \lambda^k, x \rangle,$$

setting $0 \times \infty = \infty$. When $\omega^k \neq 0$, this is the same oracle as always.

Application to the aggregative problem

We turn back to the aggregative problem. Recall its Lagrangian (with $g^*(\lambda) = \langle \lambda, b \rangle$):

$$L(x, \lambda) = \frac{1}{N} \sum_{i=1}^N \left(F_i(x_i) + \langle A_i^\top \lambda, x_i \rangle \right) - \langle \lambda, b \rangle$$

and dual cost $D(\lambda) = \inf_x L(x, \lambda)$.

Lemma

For any $\lambda \in \mathbb{R}^m$, it holds that

$$\partial d(\lambda) = \left\{ y = \frac{1}{N} \sum_{i=1}^N A_i x_i - b \mid x_i \in \operatorname{argmin} F_i(\cdot) + \langle A_i^\top \lambda, \cdot \rangle \right\}$$

Application

Algorithm 7: Dual subgradient ascent for $(\mathfrak{P})$

Initialize: $\lambda^0 \in \mathbb{R}^m$;

for $k = 0, 1, \dots, K - 1$ **do**

for $i = 1, \dots, N$ **do**

 Find $s_i^k \in \operatorname{argmin} F_i(\cdot) + \langle A_i^\top \lambda^k, \cdot \rangle$;

end

 Choose $\theta_k > 0$;

 Set $\lambda^{k+1} = \lambda^k + \theta_k (\frac{1}{N} \sum_{i=1}^N A_i s_i^k - b)$;

end

Set $x^K = \frac{1}{K} \sum_{k=0}^{K-1} s^k$.

Lemma

Choose $\theta_k = M/\sqrt{K}$, for all $k = 0, \dots, K - 1$. An output of precision of ε is obtained with a complexity of $\mathcal{O}\left(\frac{N}{\varepsilon^2}\right)$.

Randomization

The dual subgradient ascent can be randomized... but not the final primal reconstruction (why?).

Algorithm 8: Stochastic Dual Subgradient for $(\mathfrak{P})$

Initialize: $\lambda^0 \in \mathbb{R}^m$;

for $k = 0, 1, \dots, K - 1$ **do**

 Choose uniformly at random $I_k \in \{1, \dots, N\}$;

 Find $s_{I_k}^k \in \operatorname{argmin} F_{I_k}(\cdot) + \langle A_{I_k}^\top, \cdot \rangle$;

 Choose $\theta_k > 0$;

 Set $\lambda^{k+1} = \lambda^k + \theta_k (A_{I_k} s_{I_k}^k - b)$;

end

The two-stage approach

Summary of our approach:

- we use the stochastic dual subgradient ascent to compute an approximation of the value of the problem
- the primal reconstruction is done with the block-coordinate Frank-Wolfe algorithm, applied to the auxiliary problem.

Theorem

The two-stage algorithm generates an output of precision $\varepsilon > 0$ (in expectation) with a complexity of $\mathcal{O}\left(\frac{1}{\varepsilon^2} + \frac{N}{\varepsilon}\right)$.

Remark: If the block-coordinate Frank-Wolfe algorithm is properly initialized, the complexity is improved to $\mathcal{O}\left(\frac{1}{\varepsilon^2} + \frac{N}{\varepsilon^{2/3}}\right)$.

Two-stage algorithm

What if the F_i are non-convex? The **relaxation approach** remains valid in this context:

- relax the problem with probability measures
- solve the relaxed problem with the two-stage algorithm
- all subproblems produce a solution in the form of a Dirac
- recover a solution to the original problem by sampling, this induces an error term of $\mathcal{O}(\frac{1}{\sqrt{N}})$.

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Wrapping up everything!

	Smooth coupling	Coupling through an equality
Complexity	$\mathcal{O}\left(\frac{1}{\varepsilon} + \frac{N}{\sqrt{\varepsilon}}\right)$	$\mathcal{O}\left(\frac{1}{\varepsilon^2} + \frac{N}{\varepsilon}\right)$
Relaxation gap in non-convex case	$\mathcal{O}\left(\frac{1}{N}\right)$	$\mathcal{O}\left(\frac{1}{\sqrt{N}}\right)$

1 Introduction

2 A recap on convex analysis

- Main tools
- The Fenchel-Rockafellar duality theorem

3 Linear-oracle methods

- Frank-Wolfe algorithm
- The non-convex case
- Dual Subgradient

4 Proximal methods

Intention

We review the important **proximal methods**, from the augmented Lagrangian to the Chambolle-Pock algorithm.

Which of the them allow to “decompose” our aggregative problem?

We will assume that the prox of the function F_i is available.

Prox operator

Consider the minimization of $f \in \Gamma_0(\mathbb{R}^n)$ with the subgradient algorithm. The successor point x_+ of a given point x is defined as

$$x_+ = x - \tau\lambda, \quad \text{where } \lambda \in \partial f(x).$$

This is an explicit update of x . An implicit update writes:

$$x_+ = x - \tau\lambda, \quad \text{where } \lambda \in \partial f(x_+).$$

Lemma

Let $f \in \Gamma_0(\mathbb{R}^n)$, let $x \in \mathbb{R}^n$ and let $\tau > 0$. There exists a unique pair $(x, \lambda) \in \mathbb{R}^n \times \mathbb{R}^n$ such that $x_+ = x - \tau\lambda$ and $\lambda \in \partial f(x_+)$. Moreover, x_+ is characterized as the unique solution to the problem

$$\inf_{y \in \mathbb{R}^n} \tau f(y) + \frac{1}{2} \|y - x\|^2 \iff \inf_{y \in \mathbb{R}^n} f(y) + \frac{1}{2\tau} \|y - x\|^2.$$

It is denoted $\text{prox}_{\tau f}(x)$.

Two properties

Lemma

Let f_1 and $f_2 \in \Gamma_0(\mathbb{R}^n)$. Let $f: (x_1, x_2) \mapsto f_1(x_1) + f_2(x_2)$. Then,

$$\text{prox}_{\tau f}(x_1, x_2) = (\text{prox}_{\tau f_1}(x_1), \text{prox}_{\tau f_2}(x_2)).$$

Proposition (Moreau identity)

Let $f \in \Gamma_0(\mathbb{R}^n)$, let $x \in \mathbb{R}^n$ and let $\tau > 0$. Then it holds that

$$\text{prox}_{\tau f}(x) + \tau \text{prox}_{f^*/\tau}(x/\tau) = x.$$

Augmented Lagrangian

Consider a new abstract problem:

$$\inf_{x \in \mathbb{R}^n} h(x), \quad \text{subject to } Bx = c. \quad (\text{p}')$$

The Lagrangian writes:

$$\ell(x, \lambda) = h(x) + \langle \lambda, Bx - c \rangle.$$

Any pair $(\bar{x}, \bar{\lambda})$ is primal-dual optimal if and only if $-B^\top \bar{\lambda} \in \partial f(\bar{x})$ and $B\bar{x} = c$.

We define the **augmented Lagrangian** as

$$\ell_\tau(x, \lambda) = \ell(x, \lambda) + \frac{1}{2\tau} \|Bx - c\|^2.$$

Augmented Lagrangian

Lemma

Let $\bar{\lambda}$ be a dual solution. Then for any $x \in \mathbb{R}^n$, it holds that

x is a solution to (p') $\iff x$ minimizes $\ell_\tau(x, \lambda)$.

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Augmented Lagrangian

Let us come back to $\inf_{x \in \mathbb{R}^n} f(x) + g(Ax)$. We re-write it as

$$\inf_{(x,y) \in \mathbb{R}^n \times \mathbb{R}^m} f(x) + g(y), \quad \text{subject to: } Ax - y = 0. \quad (\text{p})$$

The associated **augmented Lagrangian** writes:

$$\ell_\tau(x, y, \lambda) = f(x) + g(y) + \langle \lambda, Ax - y \rangle + \frac{1}{2\tau} \|Ax - y\|^2.$$

For the aggregative problem, we obtain:

$$\begin{aligned} L_\tau(x, y, \lambda) = & \frac{1}{N} \sum_{i=1}^N F_i(x_i) + G(y) + \left\langle \lambda, \frac{1}{N} \sum_{i=1}^N A_i x_i - y \right\rangle \\ & + \frac{1}{2\tau} \left\| \frac{1}{N} \sum_{i=1}^N A_i x_i - y \right\|^2. \end{aligned}$$

Problem: it is **not separable** with respect to the variable x_i !

ADMM

A common approach to deal with the difficulty of minimizing the augmented Lagrangian consists in minimizing ℓ_τ with respect to x , then with respect to y , and doing the dual update.

Algorithm 10: Alternating Direction Method of Multipliers

Initialize: $\lambda^0 \in \mathbb{R}^m$ and y^0 ;

for $k = 0, 1, \dots, \dots$ **do**

 Compute $x^{k+1} \in \operatorname{argmin} \ell_\tau(\cdot, y^k, \lambda^k)$;

 Compute $y^{k+1} \in \operatorname{argmin} \ell_\tau(x^{k+1}, \cdot, \lambda^k)$;

 Set $\lambda^{k+1} = \lambda^k + \frac{1}{\tau}(Ax^{k+1} - y^{k+1})$;

end

But this does not fix the issue of the non-separability of ℓ_τ with respect to x for our aggregative problem.

ADMM

Consider the problem (p) with $A = \text{Id}$, i.e.:

$$\inf_{x \in \mathbb{R}^n} f(x) + g(y), \quad \text{subject to: } x - y = 0.$$

The augmented Lagrangian writes

$$\ell_\tau(x, y, \lambda) = f(x) + g(y) + \langle \lambda, x - y \rangle + \frac{1}{2} \|y - x\|^2.$$

Algorithm 11: ADMM

Initialize: $\lambda^0 \in \mathbb{R}^m$ and y^0 ;

for $k = 0, 1, \dots, \dots$ **do**

 Compute $x^{k+1} \in \text{prox}_{\tau f}(y^k - \tau \lambda^k)$;

 Compute $y^{k+1} \in \text{prox}_{\tau g}(x^k + \tau \lambda^k)$;

 Set $\lambda^{k+1} = \lambda^k + \frac{1}{\tau}(x^{k+1} - y^{k+1})$;

end

From ADMM to DRS

Lemma

Let $(x^k)_{k \in \mathbb{N}}$, $(y^k)_{k \in \mathbb{N}}$ and $(\lambda^k)_{k \in \mathbb{N}}$ be generated by ADMM. Define

$$\begin{aligned} s^k &= x^{k+1} + \tau \lambda^k \\ u^{k+1} &= y^{k+1} \\ v^{k+1} &= x^{k+2}. \end{aligned}$$

Then the sequences $(s^k)_{k \in \mathbb{N}}$, $(u^k)_{k \in \mathbb{N}}$, and $(v^k)_{k \in \mathbb{N}}$ coincide with the output of the Douglas-Rachford splitting:

$$\begin{aligned} u^{k+1} &= \text{prox}_{\tau g}(s^k) \\ v^{k+1} &= \text{prox}_{\tau f}(2u^{k+1} - s^k) \\ s^{k+1} &= s^k + v^{k+1} - u^{k+1}. \end{aligned}$$

From DRS to Chambolle-Pock

Lemma

Let $(s^k)_{k \in \mathbb{N}}$, $(u^k)_{k \in \mathbb{N}}$, $(v^k)_{k \in \mathbb{N}}$ be as in the previous lemma. Define

$$x^k = u^k$$

$$\lambda^{k+1} = \frac{1}{\tau}(u^k - s^k)$$

$$\bar{x}^{k+1} = 2x^{k+1} - x^k.$$

Then the sequences $(x^k)_{k \in \mathbb{N}}$, $(\bar{x}^k)_{k \in \mathbb{N}}$, $(\lambda^k)_{k \in \mathbb{N}}$ coincide with the output of the Chambolle-Pock algorithm:

$$\lambda^{k+1} = \text{prox}_{g^*/\tau}(\lambda^k + \frac{1}{\tau}\bar{x}^k)$$

$$x^{k+1} = \text{prox}_{\tau f}(x^k - \tau\lambda^k)$$

$$\bar{x}^{k+1} = 2x^{k+1} - x^k.$$

Chambolle-Pock algorithm

The Chambolle-Pock algorithm (a.k.a. Primal-Dual Hybrid Gradient algorithm) allows to solve our abstract problem:

$$\inf_{x \in \mathbb{R}^n} f(x) + g(Ax), \quad (\text{p})$$

for $f \in \Gamma_0(\mathbb{R}^n)$ and $g \in \Gamma_0(\mathbb{R}^m)$.

Advantages:

- Only the **knowledge of the prox operators of f and g** is required.
- **No regularity assumption** is required! Improved rates are obtained if g is smooth.
- For $f = \frac{1}{N} \sum_{i=1}^N F_i$, the computation of the prox F boils down to computing the prox's of $F_1, \dots, F_N$.
→ Allows the decomposition of $(\mathfrak{P})!$

Chambolle-Pock algorithm

Algorithm 12: Chambolle-Pock algorithm

Parameters: $\tau > 0$ and $\sigma > 0$ such that $\tau\sigma\|A\|^2 < 1$;

Initialize: $x^0, \lambda^0, \bar{x}^0 = x^0$;

for $k = 0, \dots, K$ **do**

 Compute $\lambda^{k+1} = \text{prox}_{\sigma g^*}(\lambda^k + \sigma A \bar{x}^k)$;

 Compute $x^{k+1} = \text{prox}_{\tau f}(x^k - \tau A^* \lambda^{k+1})$;

 Set $\bar{x}^{k+1} = 2x^{k+1} - x^k$;

end

Return: $\bar{x}^K = \frac{1}{K} \sum_{k=1}^K x^k$ and $\bar{\lambda}^K = \frac{1}{K} \sum_{k=1}^K \lambda^k$.

Convergence

Theorem

There exists a constant C such that for any $K \geq 0$,

$$\mathcal{G}(\bar{x}^k, \bar{\lambda}^k) \leq \frac{C}{K},$$

where $\mathcal{G}$ denotes the restricted primal-dual gap.

Comments.

- Concerning the restricted primal-dual gap:
 - If g is smooth, then $\mathcal{G}$ is an upper bound of the primal gap.
 - If g is the indicator function of b , then $\mathcal{G}$ is an upper bound of the optimality measure opt introduced before.
- When g is smooth, a modification of the algorithm yielding a rate in $\mathcal{O}(C/K^2)$ is possible.

Application to the aggregative problem

The Chambolle-Pock algorithm almost directly applies to the aggregative problem, using the **weighted** norm

$$\|(x_1, \dots, x_N)\|_w := \frac{1}{N} \sum_{i=1}^N \|x_i\|^2.$$

The following **complexities** are reached:

	Smooth coupling	Coupling through an equality
Linear oracle	$\mathcal{O}\left(\frac{1}{\varepsilon} + \frac{N}{\sqrt{\varepsilon}}\right)$	$\mathcal{O}\left(\frac{1}{\varepsilon^2} + \frac{N}{\varepsilon}\right)$
Prox oracle	$\mathcal{O}\left(\frac{N}{\sqrt{\varepsilon}}\right)$	$\mathcal{O}\left(\frac{N}{\varepsilon}\right)$

Last comments

Last comments...

- The application of Chambolle Pock to the relaxed problem, in the non-convex case, is “delicate”.
- A randomization of Chambolle-Pock exists, but does not yield a theoretical improvement of the complexity.

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Thank you for your attention!